

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA1506
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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I _____M.NITHIYASHREE-192421101_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____M.NITHIYASHREE_____

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

ANSWERS

1. Compare Type-1 and Type-2 Hypervisors for a Cloud Datacenter Hosting Mission-Critical Applications. Analyze Performance, Security, and Manageability, then Justify the Better Choice.

Comparison

Feature	Type-1 Hypervisor	Type-2 Hypervisor
Installation	Runs directly on hardware (bare metal)	Runs on top of a host operating system
Performance	High performance with low overhead	Lower performance due to host OS overhead
Security	Strong isolation and smaller attack surface	More vulnerable because host OS can be attacked

Manageability	Supports centralized management and live migration	Limited enterprise management features
Examples	VMware ESXi, Microsoft Hyper-V, Xen	VMware Workstation, Oracle VirtualBox

Analysis

Mission-critical applications require high availability, fast response time, and strong security. Type-1 hypervisors interact directly with the hardware, reducing resource overhead and improving CPU and memory utilization. They also provide better VM isolation and enterprise features such as live migration, snapshots, clustering, and fault tolerance.

Type-2 hypervisors depend on the host operating system. If the host OS fails or is compromised, all virtual machines may also be affected. This makes them less suitable for production cloud environments.

Justification

Type-1 Hypervisor is the better choice because it offers:

- Better performance through direct hardware access.
- Stronger security with improved VM isolation.
- High availability using clustering and live migration.
- Easier management through centralized administration.

2. A Cloud Provider Must Isolate Workloads Belonging to Finance, Healthcare, and Student Portals on the Same Hardware. Analyze How Virtualization Architecture Supports Isolation and Identify Two Major Attack Surfaces.

Virtualization-Based Isolation

Virtualization allows multiple virtual machines to run independently on the same physical server. Each VM has its own operating system, CPU allocation, memory, storage, and virtual network.

For example:

- **Finance VM** stores banking transactions.
- **Healthcare VM** stores patient records.

- **Student Portal VM** handles academic information.

The hypervisor ensures that one VM cannot directly access another VM's resources.

Isolation Techniques

- Memory isolation
- CPU scheduling
- Virtual storage separation
- Virtual LANs (VLANs)
- Access control policies
- Network segmentation

Major Attack Surfaces

1. Hypervisor Attack

If attackers exploit vulnerabilities in the hypervisor, they may gain control over multiple VMs running on the server.

2. VM Escape Attack

A compromised virtual machine attempts to escape its isolated environment and access the host system or neighboring VMs.

Security Measures

- Regular hypervisor patching
 - Secure boot
 - Multi-factor authentication
 - Encryption
 - Network firewalls
 - Intrusion Detection Systems (IDS)
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3. Study the Following VM Host Utilization Snapshot and Recommend Corrective Actions:

Given Data

- CPU Utilization = **92%**
- Memory Utilization = **88%**

- Storage I/O Latency = **35 ms**
- Average VM Boot Time = **170 seconds**

Analysis

CPU Bottleneck

CPU usage of **92%** indicates processor saturation. Applications compete for CPU resources, causing slow execution and delayed VM startup.

Corrective Actions

- Add more CPU cores.
 - Migrate some VMs to another host.
 - Enable CPU load balancing.
 - Increase auto-scaling.
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Memory Bottleneck

Memory utilization of **88%** is close to maximum capacity. This may cause swapping, reducing overall performance.

Corrective Actions

- Increase physical RAM.
 - Use memory optimization techniques.
 - Reduce memory overcommitment.
 - Optimize VM memory allocation.
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Storage Bottleneck

Storage latency of **35 ms** is high. Slow disk operations increase VM startup time and application response time.

Corrective Actions

- Replace HDDs with SSDs or NVMe drives.
- Enable storage caching.
- Upgrade SAN/NAS storage.
- Use storage tiering.

VM Boot Time

A boot time of **170 seconds** is higher than expected.

Reasons

- High CPU load
- Memory shortage
- Slow storage access

Overall Recommendation

- Scale compute resources.
 - Increase RAM.
 - Upgrade storage.
 - Implement auto-scaling.
 - Use load balancing to distribute VM workloads.
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4. Analyze How Software Defined Datacenter (SDDC) Architecture Improves Agility Compared with a Traditional Datacenter. Support Your Answer with Compute, Storage, and Network Virtualization Considerations.

Traditional Datacenter

Traditional datacenters require manual configuration of servers, storage, and networking. Resource allocation is slow and hardware-dependent.

Software Defined Datacenter

An SDDC virtualizes all infrastructure components and manages them through software. Resources are automatically provisioned, monitored, and scaled.

Comparison

Aspect	Traditional Datacenter	Software Defined Datacenter
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Resource Allocation	Manual	Automated
Scalability	Limited	Dynamic
Deployment Time	Hours or days	Minutes
Management	Hardware-based	Software-based
Cost	High	Lower due to automation

Compute Virtualization

Physical servers are converted into multiple virtual machines, enabling efficient resource sharing and rapid provisioning.

Storage Virtualization

Different storage devices are combined into a single virtual storage pool, allowing flexible allocation and improved utilization.

Network Virtualization

Software Defined Networking (SDN) creates virtual networks independent of physical hardware, enabling quick configuration, traffic management, and security enforcement.

Benefits

- Faster deployment
 - Better scalability
 - Higher resource utilization
 - Automated management
 - Reduced operational cost
 - Improved disaster recovery
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5. A Multi-Cloud Federation Project Fails Due to Identity Mismatches Between Providers. Analyze the Issue and Propose a Secure Identity and Privacy Design Using Federation Concepts.

Problem Analysis

Different cloud providers use different identity management systems. User identities, authentication methods, and access policies may not be compatible.

As a result:

- Users may need multiple logins.
- Authentication failures occur.
- Access permissions become inconsistent.
- Security risks increase.

Proposed Secure Identity Design

Identity Federation

Implement a centralized Identity Provider (IdP) that authenticates users once and allows secure access to multiple cloud providers.

Authentication Protocols

Use:

- SAML (Security Assertion Markup Language)
- OAuth 2.0
- OpenID Connect (OIDC)

These protocols provide secure Single Sign-On (SSO).

Privacy Protection

- Encrypt user credentials and sensitive data.
- Use Multi-Factor Authentication (MFA).
- Apply Role-Based Access Control (RBAC).
- Share only required user attributes between providers.
- Maintain audit logs for monitoring.

Architecture

User

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Identity Provider (IdP)

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|—————▶ Cloud Provider A

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|—————▶ Cloud Provider B

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|—————▶ Cloud Provider C

Benefits

- Single Sign-On (SSO)
- Consistent identity management
- Strong authentication
- Improved privacy
- Reduced administrative overhead
- Secure access across multiple cloud providers

Conclusion

Virtualization is the foundation of modern cloud computing. Type-1 hypervisors provide the best performance and security for mission-critical workloads. Virtualization architecture ensures workload isolation while minimizing security risks. Monitoring resource utilization helps identify CPU, memory, and storage bottlenecks. Software Defined Datacenters improve agility through compute, storage, and network virtualization, while identity federation enables secure and seamless access across multiple cloud providers. These technologies together enhance the scalability, security, and efficiency of cloud environments.